

GPS EMERGENCY TRACKER

With Voice-Activated Hands-Free SOS Alerts

A Smart, Web-Based Emergency Response Ecosystem for Vulnerable Individuals

Hackathon Pitch Presentation

Built with Flask | WebSockets | Web Speech API | Twilio API

The Critical Problem

The Human Vulnerability

- In physical assaults, accidents, or medical crises (e.g. cardiac arrest, falls), victims often cannot reach or operate their mobile screen.
- Elderly individuals or physically disabled users face severe cognitive or motor barriers when interacting with complex smart devices during high-stress situations.
- Every second of delay in dispatching precise GPS location data directly impacts safety outcomes and response times.

Limitations of Existing Solutions

- Standard panic apps require active device unlocking, swiping, or visual focus, which is impossible during direct threats.
- Most safety tracking tools rely on heavy server polling, leading to high latency and battery drain instead of instant notification updates.
- Critical alert notifications are often limited to single-channel notifications (like email only) that emergency contacts easily miss.

The Smart Solution

An intelligent, hands-free emergency tracking and alerting system.

□ **Hands-Free Voice Activation:** Integrates Web Speech API directly in the browser, triggering an instant emergency response upon speaking standard keywords ('Help me', 'SOS', 'Emergency') without touching the screen.

📶 **Proactive Safety Sensors:** Detects falls and periods of abnormal inactivity automatically, sending immediate alerts even if the user is unconscious or incapacitated.

📍 **Real-Time Spatial Awareness:** Continuous live tracking utilizing browser geolocation rendered on interactive Leaflet.js maps, paired with custom geofencing safety boundaries.

📞 **Multi-Channel Urgent Broadcasts:** Triggers simultaneous voice call notifications (via Twilio automated text-to-speech), Twilio SMS alerts, and detailed HTML emergency emails to all registered contacts.

Core Product Features

Voice Assistant Trigger

Triggers panic alarms and emergency dispatch immediately using natural language commands. Designed for situations where a device cannot be reached physically.

Real-Time Alert Engine

Instantly fires Twilio SMS messages, initiates automated interactive voice calls, and distributes rich SMTP emails detailing the user's name and exact GPS link.

Live Mapping & Geofencing

Interactive Leaflet.js spatial views. Allows administrators to establish virtual safe zones. Triggers warning alarms automatically if a user crosses boundaries.

Fall & Motion Detection

Leverages device accelerometer/gyroscope simulations and motion patterns. Detects sudden falls or extended inactivity, dispatching automatic SOS alerts.

Technical Architecture

□ Front-End client

- **HTML5 & Vanilla CSS:**
Modern, responsive glassmorphism UI.
- **Leaflet.js:**
Lightweight interactive mapping library.
- **Web Speech API:**
Low-latency local voice speech-to-text recognition.
- **Socket.IO Client:**
Real-time, bi-directional server connectivity.

⚙️ Back-End engine

- **Flask (Python):**
Lightweight WSGI application controller.
- **Flask-SocketIO:**
WebSocket orchestration for real-time client updates.
- **SQLite & SQLAlchemy:**
Structured, high-performance local relational DB.
- **Background Threads:**
Handles non-blocking notification tasks.

📞 Third-Party integrations

- **Twilio SMS API:**
Low-latency mobile distress texts.
- **Twilio Voice Call API:**
Text-to-speech phone call broadcasting.
- **Gmail SMTP:**
Secure HTML alert delivery (App Passwords).
- **HTML5 Geolocation:**
High-accuracy coordinates API.

Real-World Impact



Elder Care & Independent Living

Automatically detects fall events or prolonged lack of movement in senior citizens, triggering immediate automated voice calls and emails to caregivers.



Personal Safety & Anti-Assault

Allows users to trigger immediate silent distress commands ('Save me', 'Call police') under threat when dialing manually is unsafe or impossible.



High-Risk Industrial Environments

Protects lone workers in mines, factories, or construction zones. Relies on active movement tracking and boundary/geofence alerts for security teams.

Future Roadmap

PHASE 1

IoT Wearables

Develop integrations for smart bands and watch accelerators (e.g. WearOS) to detect falls and record heart rate metrics autonomously.

PHASE 2

Native Apps

Release native iOS and Android versions using Flutter/React Native, enabling background GPS reporting and offline SMS fall sensing.

PHASE 3

AI Diagnostics

Integrate local machine learning models for acoustic panic detection, voice frequency stress analysis, and behavioral pattern anomaly detection.

PHASE 4

Mesh Networks

Deploy Bluetooth/LoRa WAN mesh fallback networks to enable message delivery even in regions without cellular coverage or internet.

Try the Project Live

Thank You!

Ready for live demonstration

GitHub Repository:

<https://github.com/balu738697/Emergency-gps-tracking>

Open run.bat locally to launch the portal instantly!